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Adaptive Vibration Dampening System for Small-Scale Urban Wind Turbines

ABSTRACT

A vibration dampening system for small-scale wind turbines is disclosed, aimed at mitigating noise and structural fatigue caused by operational vibrations. The system comprises multiple tunable dampers strategically positioned on the turbine tower and nacelle.

Each damper utilizes magnetorheological (MR) fluid whose viscosity can be altered by a magnetic field, thus enabling dynamic adjustments to the damping characteristics. A control unit, receiving real-time wind speed data and optional vibration feedback, governs the applied magnetic fields to optimize damping across various operating conditions. The system incorporates a self-learning algorithm to refine its response over time, ensuring adaptability to site-specific wind patterns and turbine characteristics.

BACKGROUND

Small-scale wind turbines offer an attractive renewable energy solution for urban environments. However, their operation generates vibrations that can lead to excessive noise, decreased efficiency, and potentially structural damage over time. Traditional passive damping systems often fall short in addressing the wide range of vibrational frequencies and amplitudes encountered in urban wind conditions.

SUMMARY

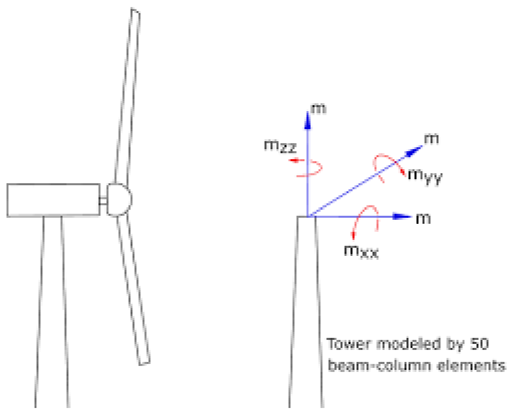
The present invention introduces an adaptive vibration dampening system specifically tailored for small-scale urban wind turbines. The system employs multiple dampers with MR fluid, enabling real-time adjustments to the damping force in response to varying wind speeds and vibration levels.

A control unit monitors wind speed via a dedicated sensor and can optionally receive data from an accelerometer mounted on the turbine. Using a proprietary algorithm, it determines the optimal viscosity for each MR damper, effectively altering the damping coefficient to match the prevailing conditions.

The system further incorporates a self-learning capability, whereby the control algorithm refines its parameters over time based on accumulated data. This ensures that the dampening system continuously adapts to the specific wind patterns and turbine dynamics at its installation site, maximizing its effectiveness.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1: Schematic diagram of the adaptive vibration dampening system integrated into a small-scale wind turbine.

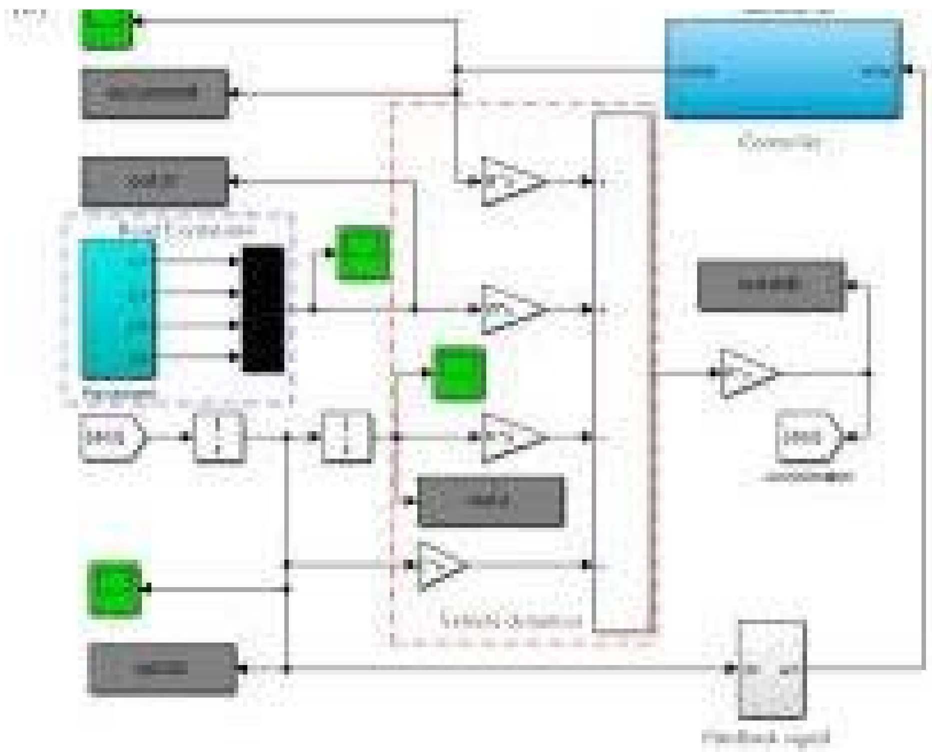


schematic diagram showing a wind turbine with dampers on the tower and nacelle, connected to a control unit, wind speed sensor, and optional accelerometer.

FIG. 2: Detailed view of a single MR damper with its associated electromagnet.

closeup view of a damper with a cylinder, piston, electromagnet, and MR fluid.

FIG. 3: Flowchart illustrating the control algorithm for adjusting damper viscosity.



flowchart showing the control algorithm with inputs from wind speed sensor and accelerometer, calculations, and outputs to electromagnets.

DETAILED DESCRIPTION

1. **System Components:**
    - Multiple tunable dampers utilizing MR fluid technology.
    - Wind speed sensor.
    - Optional accelerometer for vibration feedback.
    - Control unit with integrated microprocessor and power electronics.
2. **Damper Operation:**
    - Each damper consists of a cylinder filled with MR fluid, a piston, and an electromagnet.
    - The applied magnetic field alters the fluid viscosity, thereby changing the damping force.
    - Dampers are strategically placed on the tower and nacelle to mitigate vibrations at their sources.
3. **Control Algorithm:**
    - Wind speed data (and optionally vibration data) are continuously fed into the control unit.
    - The algorithm determines the optimal viscosity for each damper based on pre-defined parameters and accumulated learning.
    - The control unit adjusts the current to the electromagnets to achieve the desired viscosity in each damper.
4. **Self-Learning:**

- The control algorithm includes a feedback loop that modifies its parameters based on the observed vibration levels.
- Over time, this self-learning process optimizes the system's response for the specific turbine and location.

CLAIMS

1. An adaptive vibration dampening system for a small-scale wind turbine, comprising:
   - Multiple tunable dampers utilizing magnetorheological (MR) fluid.
   - A wind speed sensor for providing real-time wind speed data.
   - A control unit configured to receive said wind speed data and adjust the viscosity of each MR damper to optimize damping performance.
2. The system of claim 1, further comprising an accelerometer mounted on the wind turbine for providing real-time vibration data to the control unit.
3. The system of claim 1 or 2, wherein the control unit incorporates a self-learning algorithm that modifies its parameters based on accumulated wind speed and vibration data.
4. The system of claim 1 or 2, wherein the dampers are strategically positioned on both the tower and the nacelle of the wind turbine.
5. A method for controlling the vibration dampening system of claim 1, comprising the steps of:
   - Continuously monitoring wind speed and optionally vibration data.
   - Determining the optimal viscosity for each MR damper based on said data.
   - Adjusting the current to the electromagnets of each damper to achieve the desired viscosity.
6. The method of claim 5, further comprising the step of updating the parameters of the control algorithm based on observed vibration levels over time.

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